8p12s six-step ripple investigation

Supplemental diagnostics; original registered benchmark remains FAIL.

Measured commit: 1ea9e406818b9ceafe5408e9c16acd33e9c439b0

The original 7.5 to 3.75 degree comparison changes ripple by **5.034056246 percentage points**, above the unchanged 5.0 pp limit. All shared-angle torque values remain bit-for-bit identical at every refinement.

Step (deg)	Points	Mean (Nm)	Minimum (Nm)	Ripple (%)
7.5	48	17.446357	4.767362	147.737333
3.75	96	17.469421	3.853867	152.771389
1.875	192	17.477691	3.850692	152.717268
0.9375	384	17.478492	3.691994	153.618234

What the controls establish

The two further refinements change ripple by 0.054121 and 0.900966 pp. The finer minimum is at 12.1875 electrical degrees, inside a commutation sector (+50, -50, 0 A). The coarse grid misses this trough. The normal spatial mesh and non-angle settings were held fixed.

A fresh 25 A, 3.75-degree solve exactly reproduces all 96 torque samples from the earlier 04f87a5 fine result, whose registered request was 50 A but whose slot source was halved. The corrected 50 A field therefore exposes the sampling sensitivity. No recorded waveform was rescaled.

Disposition

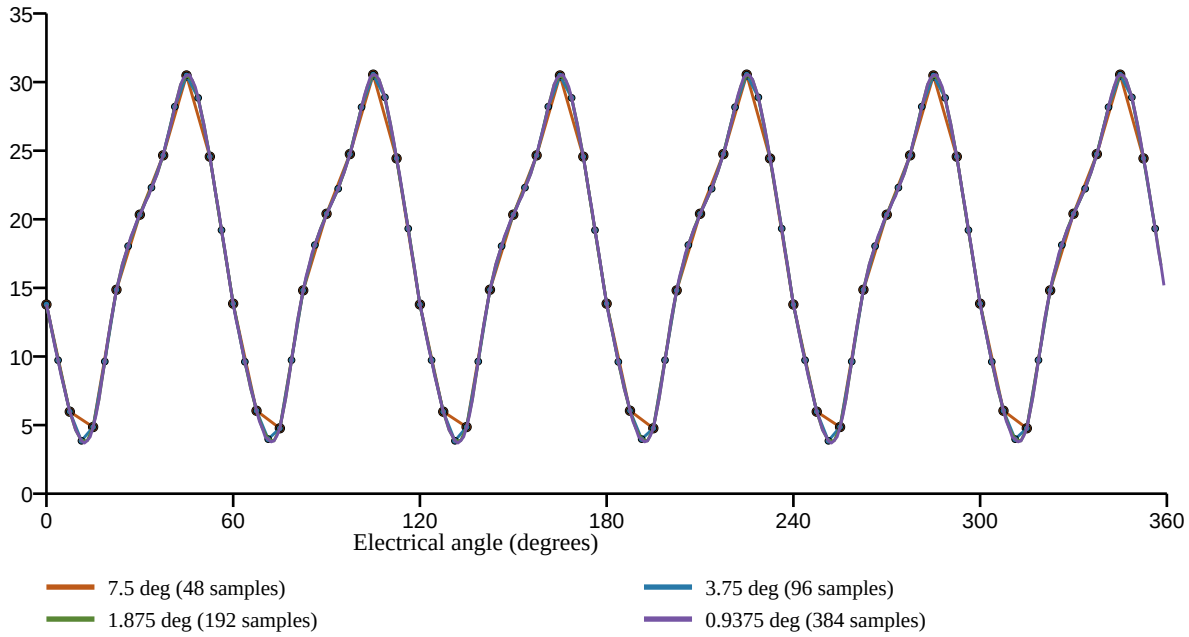
Retain the original FAIL. Qualify finer angular grids with matching FEMM samples in a future registered protocol. These diagnostics do not replace the original result, justify increasing its threshold, or establish spatial convergence.

All three supplemental jobs completed in 242.453 s wall time. Source, binary and original evidence hashes passed verification. Full request/result evidence and exact numerical values accompany this PDF.

Angular sampling plots

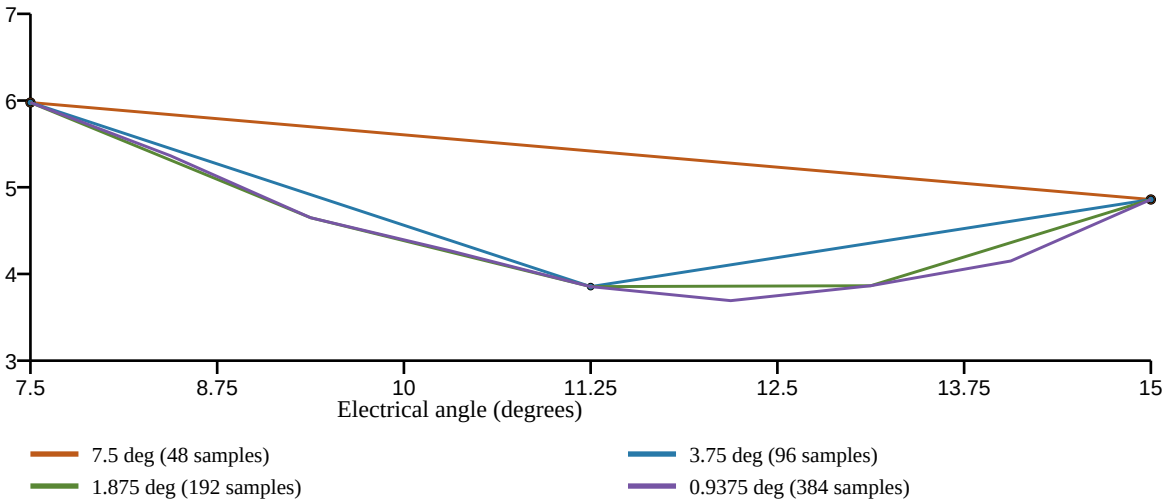
50 A torque over one full electrical cycle

Torque (Nm); straight lines connect measured samples



50 A trough: additional angles reveal a lower minimum

Torque (Nm); straight lines connect measured samples



Lines only connect samples; they do not represent additional solves or fitted extrema. The lower trough continues to shift with refinement.